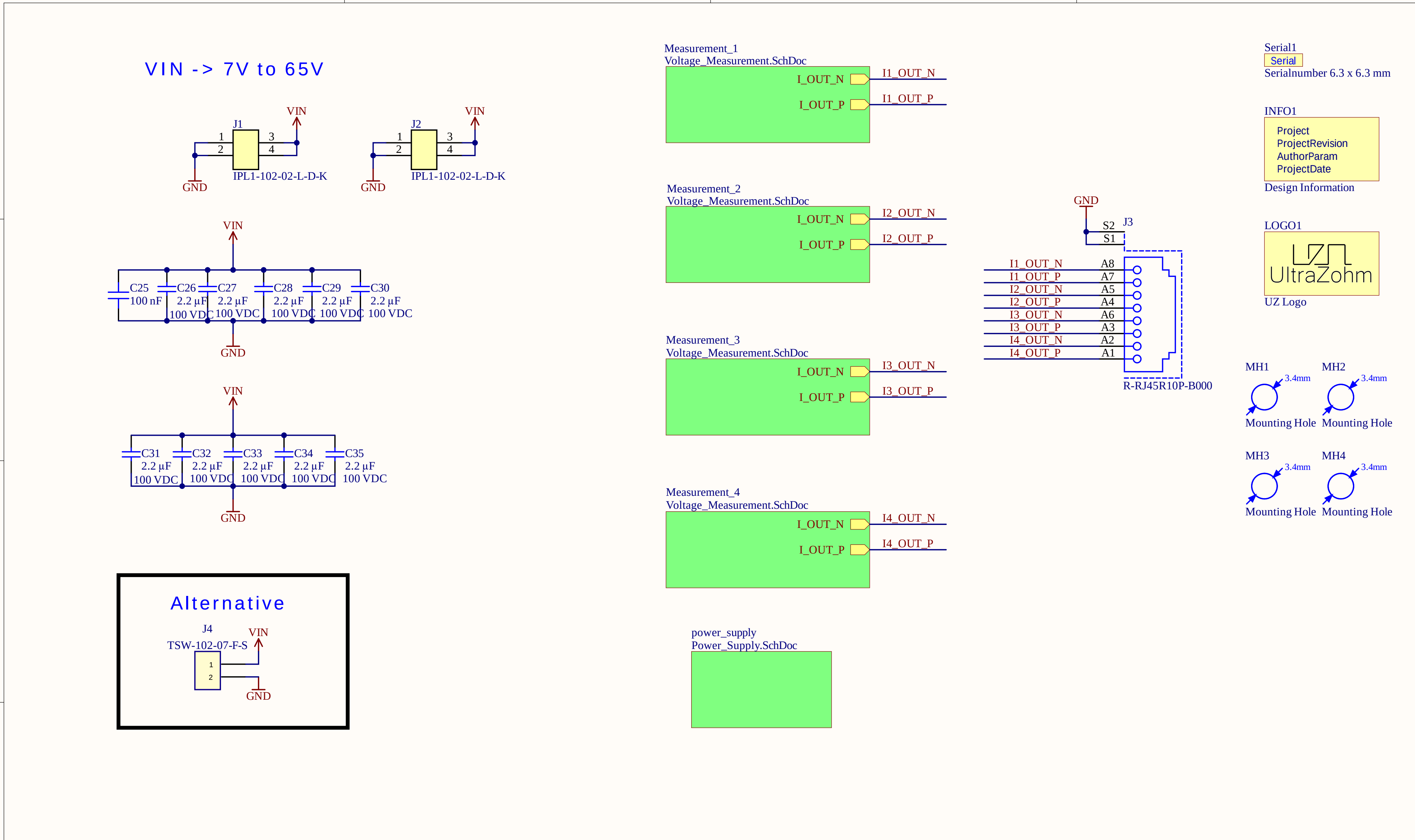

Title TopSheet.SchDoc		UltraZohm
Revision: Rev01	Design Engineer: Nina Diringer	www.ultrazohm.com
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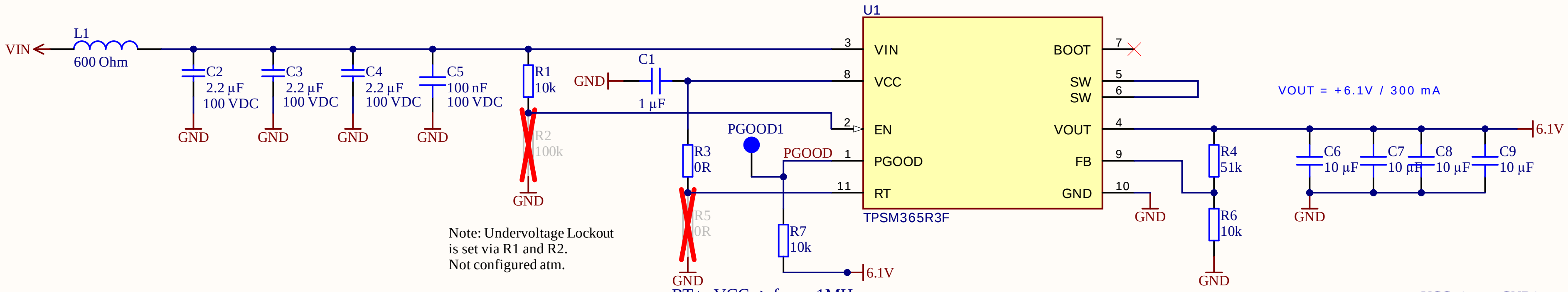
$$R_{FBT}[\mathrm{k\Omega}] = R_{FBB}[\mathrm{k\Omega}] \times \left(\frac{V_{OUT}[\mathrm{V}]}{1\ \mathrm{V}} - 1\right)$$

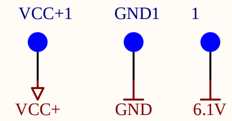


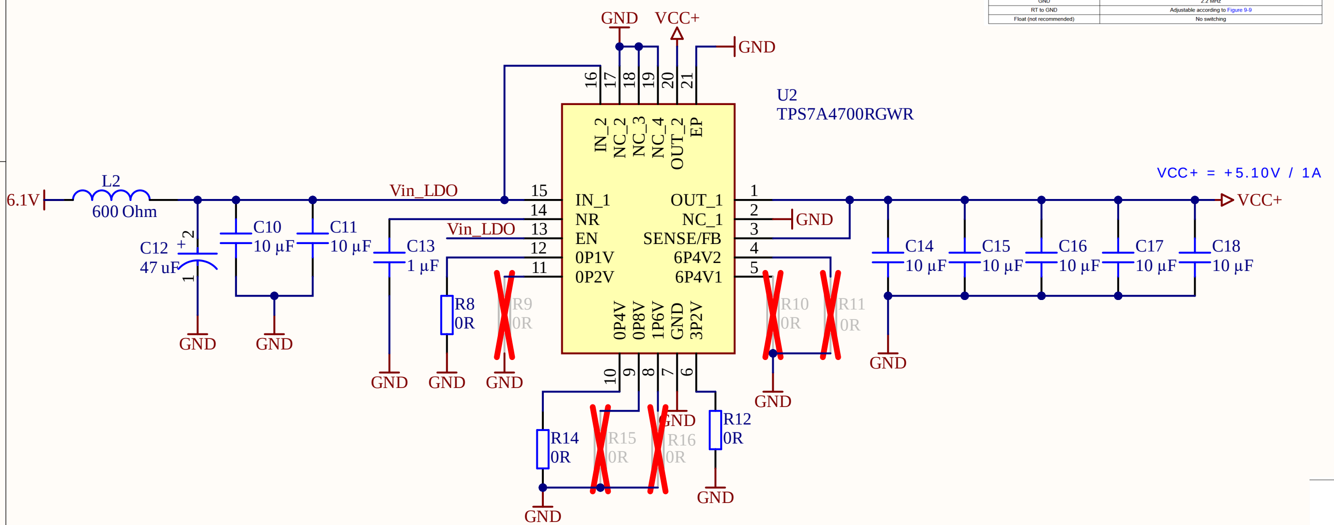
Note: Undervoltage Lockout
is set via R1 and R2.
Not configured atm.

RT to VCC -> fsw = 1MHz

Table 9-5. RT Pin Setting

RT INPUT	SWITCHING FREQUENCY
VCC	1 MHz
GND	2.2 MHz
RT to GND	Adjustable according to Figure 9-9
Float (not recommended)	No switching





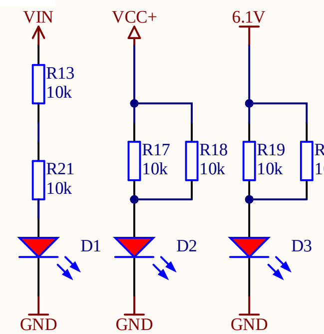
7.5.1 ANY-OUT Programmable Output Voltage

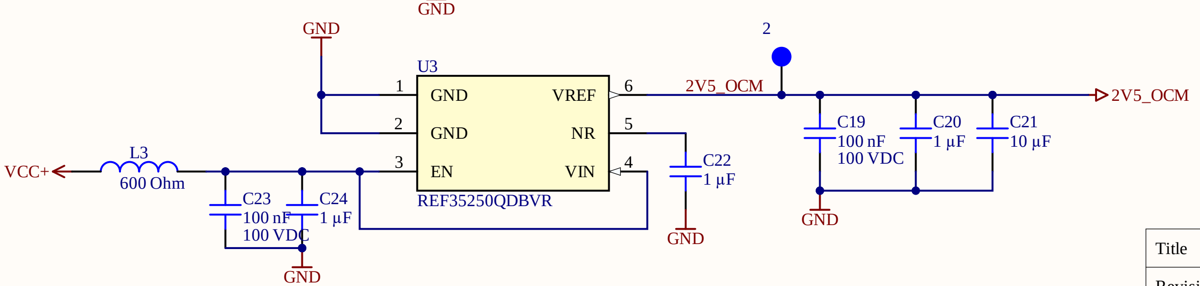
Both devices can be used in ANY-OUT mode. For ANY-OUT operation, the TPS7A4700 and TPS7A4701 do not use external resistors to set the output voltage, but use device pins 4, 5, 6, 8, 9, 10, 11, and 12 to program the regulated output voltage. Each pin is either connected to ground (active) or is left open (floating). The ANY-OUT programming is set by Equation 2 as the sum of the internal reference voltage ($V_{(REF)}$ = 1.4 V) plus the accumulated sum of the respective voltages assigned to each active pin; that is, 100 mV (pin 12), 200 mV (pin 11), 400 mV (pin 10), 800 mV (pin 9), 1.6 V (pin 8), 3.2 V (pin 6), 6.4 V (pin 5), or 6.4 V (pin 4). Table 1 summarizes these voltage values associated with each active pin setting for reference. By leaving all program pins open, or floating, the output is thereby programmed to the minimum possible output voltage equal to $V_{(REF)}$.

$$V_{OUT} = V_{REF} + (\Sigma \text{ ANY-OUT Pins to Ground}) \quad (2)$$

Table 1. ANY-OUT Programmable Output Voltage

ANY-OUT PROGRAM PINS (Active Low)	ADDITIVE OUTPUT VOLTAGE LEVEL
Pin 4 (6P4V2)	6.4 V
Pin 5 (6P4V1)	6.4 V
Pin 6 (3P2)	3.2 V
Pin 8 (1P6)	1.6 V
Pin 9 (0P8)	800 mV
Pin 10 (0P4)	400 mV
Pin 11 (0P2)	200 mV
Pin 12 (0P1)	100 mV





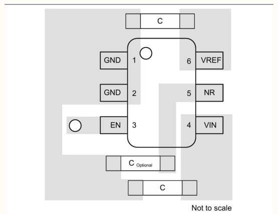


Figure 10-4. Layout Example

NR -> Noise reduction cap

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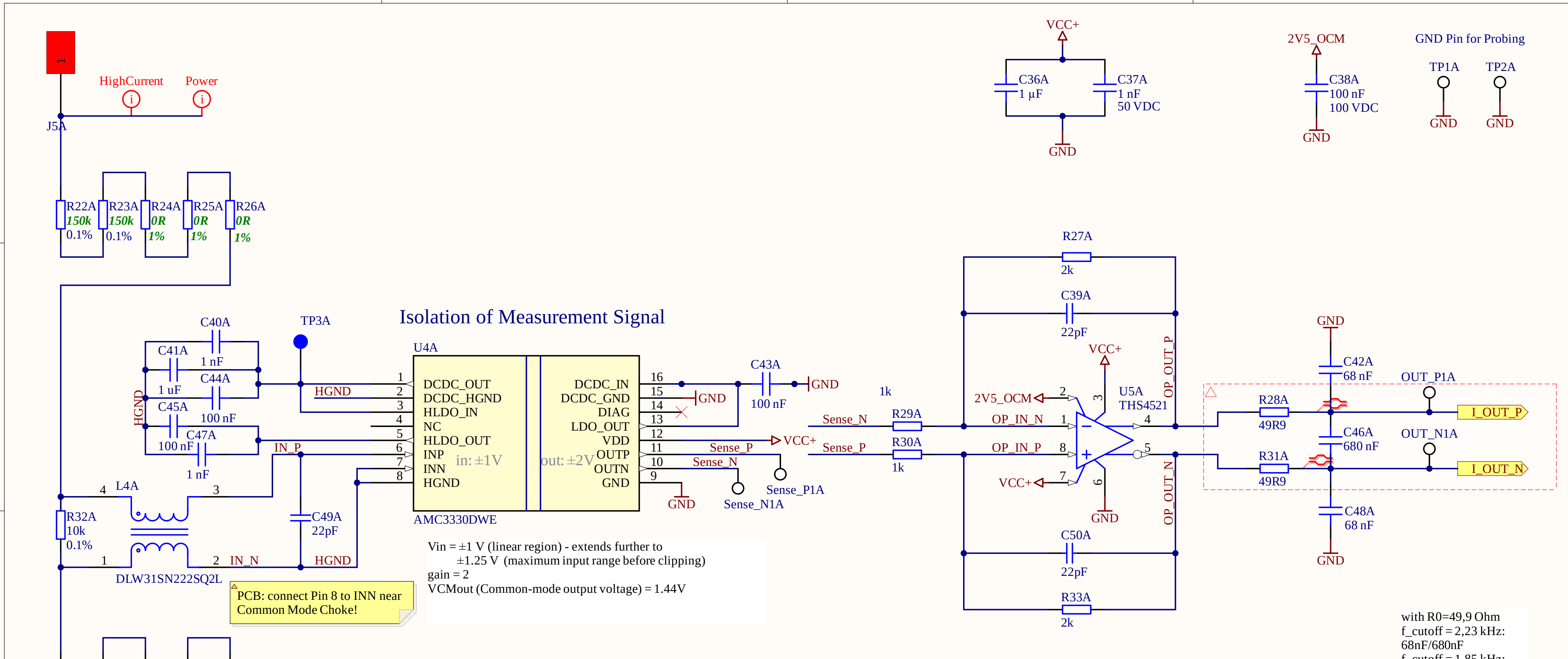
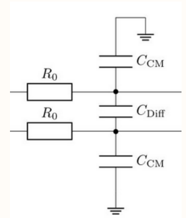


Vin = ±1 V (linear region) - extends further to
±1.25 V (maximum input range before clipping)
gain = 2
VCMout (Common-mode output voltage) = 1.44V

PCB: connect Pin 8 to INN near Common Mode Choke!

with R0=49,9 Ohm
f_cutoff = 2,23 kHz: 68nF/680nF
f_cutoff = 1,85 kHz: 82nF/820nF



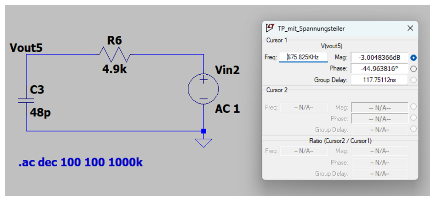
$$f_{\text{aliasing},-3\text{dB}} = \frac{1}{2\pi R_0(2C_{\text{Diff}} + C_{\text{CM}})}$$

Vdc,max = 60 V | 1000 V
I_Cross = 100uA | 100uA
R_sense_total = Vdc / I_Cross = 600k | 10 M
R_sense = 10k | 10k
R_series = 2*150k | 5*1M

V_sense (60V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 0,98V

V_sense (1000V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 1,0V

Differential Filter with voltage divider can be summarized to this:



Title	Voltage_Measurement.SchDoc
Revision: Rev01	Design Engineer: Nina Diringer
Project: uz_per_voltage_measurement.PrjPcb	
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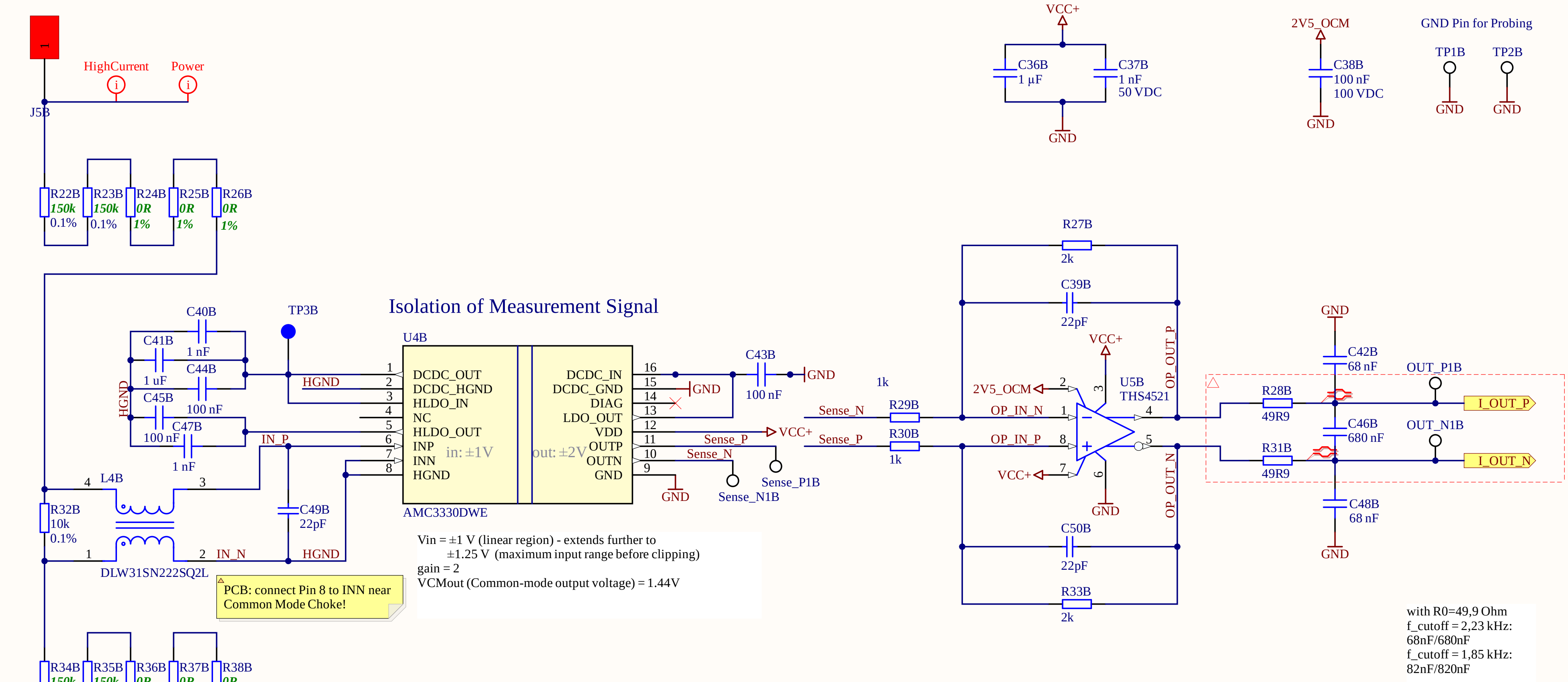
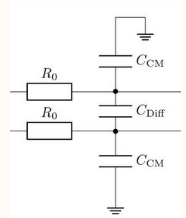
Isolation of Measurement Signal



PCB: connect Pin 8 to INN near Common Mode Choke!

Vin = ±1 V (linear region) - extends further to ±1.25 V (maximum input range before clipping)
gain = 2
VCMout (Common-mode output voltage) = 1.44V

with R0=49,9 Ohm
f_cutoff = 2,23 kHz: 68nF/680nF
f_cutoff = 1,85 kHz: 82nF/820nF



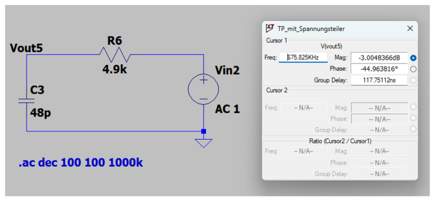
$$f_{\mathrm{aliasing},-3\mathrm{dB}} = \frac{1}{2\pi R_0(2C_{\mathrm{Diff}} + C_{\mathrm{CM}})}$$

Vdc,max = 60 V | 1000 V
I_Cross = 100uA | 100uA
R_sense_total = Vdc / I_Cross = 600k | 10 M
R_sense = 10k | 10k
R_series = 2*150k | 5*1M

V_sense (60V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 0,98V

V_sense (1000V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 1,0V

Differential Filter with voltage divider can be summarized to this:



Title	Voltage_Measurement.SchDoc
Revision: Rev01	Design Engineer: Nina Diringer
Project: uz_per_voltage_measurement.PrjPcb	Date: 30.09.2025
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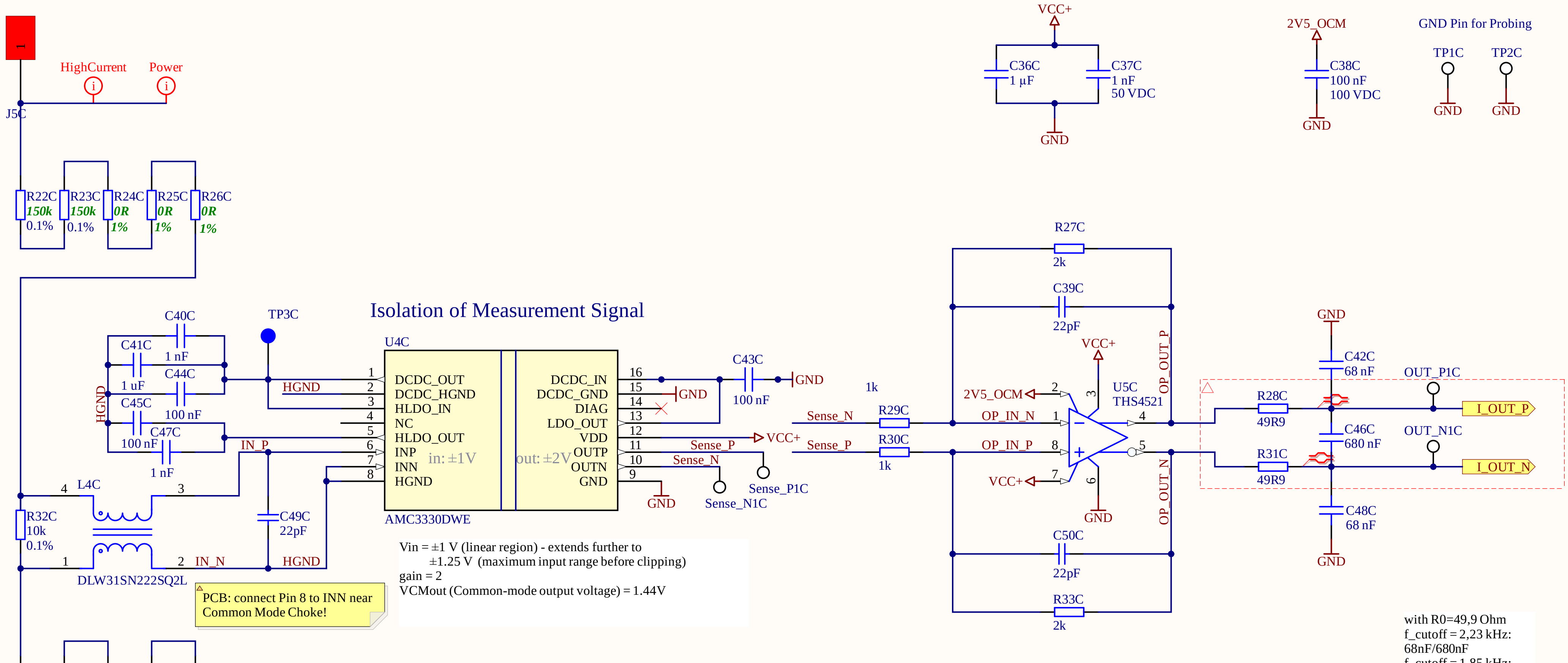
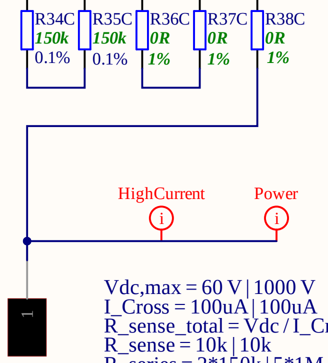




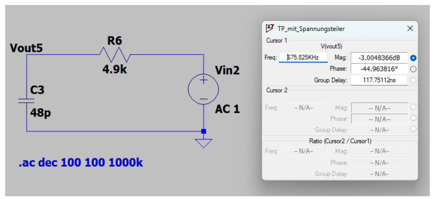
Vin = ±1 V (linear region) - extends further to
±1.25 V (maximum input range before clipping)
gain = 2
VCMout (Common-mode output voltage) = 1.44V

$V_{dc,max}$ = 60 V | 1000 V
I_{Cross} = 100uA | 100uA
R_{sense_total} = Vdc / I_Cross = 600k | 10 M
R_{sense} = 10k | 10k
R_{series} = 2*150k | 5*1M

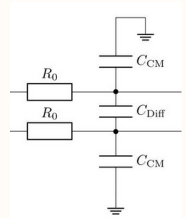
V_sense (60V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 0,98V

V_sense (1000V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 1,0V

Differential Filter with voltage divider can be summarized to this:



with R0=49,9 Ohm
f_cutoff = 2,23 kHz:
68nF/680nF
f_cutoff = 1,85 kHz:
82nF/820nF



$$f_{\text{aliasing},-3\text{dB}} = \frac{1}{2\pi R_0 (2C_{\text{Diff}} + C_{\text{CM}})}$$

Title	Voltage_Measurement.SchDoc
Revision: Rev01	Design Engineer: Nina Diringer
Project: uz_per_voltage_measurement.PrjPcb	Date: 30.09.2025
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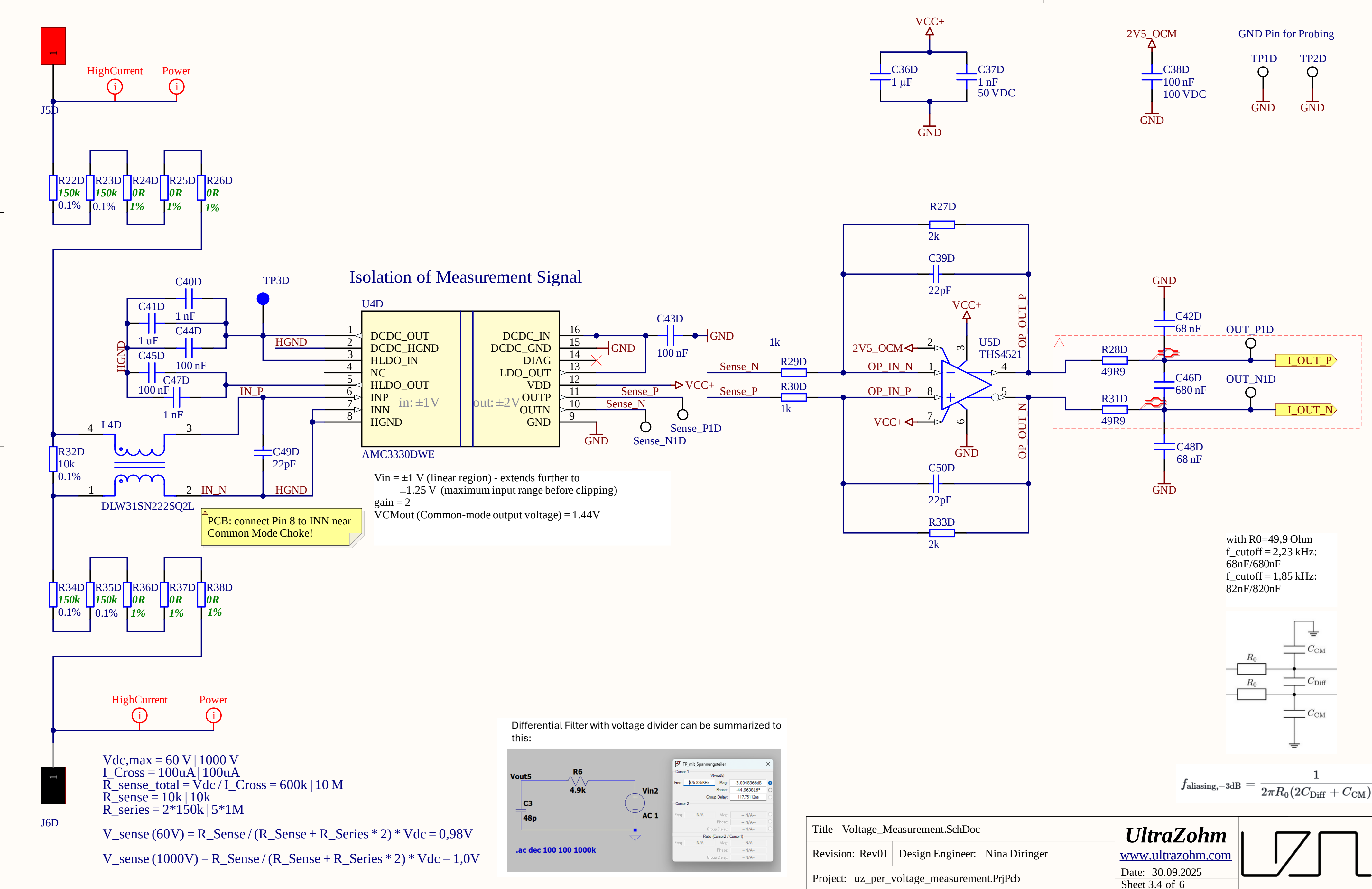
Isolation of Measurement Signal
J5D
HighCurrent
Power
R22D 150k 0.1%
R23D 150k 0.1%
R24D 0R 1%
R25D 0R 1%
R26D 0R 1%
C40D 1 nF
C41D 1 uF
C44D 100 nF
C45D 100 nF
C47D 1 nF
HGND
TP3D
U4D
AMC3330DWE
DCDC_OUT
DCDC_HGND
HLDO_IN
NC
HLDO_OUT
INP
INN
HGND
DCDC_IN
DCDC_GND
DIAG
LDO_OUT
VDD
OUTP
OUTN
GND
in: ±1V
out: ±2V
C43D 100 nF
GND
VCC+
Sense_P
Sense_N
Sense_P1D
Sense_N1D
L4D
DLW31SN222SQ2L
R32D 10k 0.1%
IN_P
IN_N
C49D 22pF
PCB: connect Pin 8 to INN near Common Mode Choke!
Vin = ±1 V (linear region) - extends further to
±1.25 V (maximum input range before clipping)
gain = 2
VCMout (Common-mode output voltage) = 1.44V
R27D 2k
C39D 22pF
R29D 1k
R30D 1k
C50D 22pF
R33D 2k
2V5_OCM
OP_IN_N
OP_IN_P
OP_OUT_P
OP_OUT_N
U5D
THS4521
R28D 49R9
R31D 49R9
C42D 68 nF
C46D 680 nF
C48D 68 nF
OUT_P1D
OUT_N1D
I_OUT_P
I_OUT_N
C36D 1 µF
C37D 1 nF 50 VDC
C38D 100 nF 100 VDC
GND Pin for Probing
TP1D
TP2D
with R0=49,9 Ohm
f_cutoff = 2,23 kHz:
68nF/680nF
f_cutoff = 1,85 kHz:
82nF/820nF
R_0
C_{CM}
C_{Diff}
$f_{\text{aliasing},-3\text{dB}} = \dfrac{1}{2\pi R_0 (2C_{\text{Diff}} + C_{\text{CM}})}$
R34D 150k 0.1%
R35D 150k 0.1%
R36D 0R 1%
R37D 0R 1%
R38D 0R 1%
J6D
Vdc,max = 60 V | 1000 V
I_Cross = 100uA | 100uA
R_sense_total = Vdc / I_Cross = 600k | 10 M
R_sense = 10k | 10k
R_series = 2*150k | 5*1M
V_sense (60V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 0,98V
V_sense (1000V) = R_Sense / (R_Sense + R_Series * 2) * Vdc = 1,0V
Differential Filter with voltage divider can be summarized to this:
Vout5
R6
4.9k
C3
48p
Vin2
AC 1
.ac dec 100 100 1000k
Title Voltage_Measurement.SchDoc
Revision: Rev01
Design Engineer: Nina Diringer
Project: uz_per_voltage_measurement.PrjPcb
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Date: 30.09.2025
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